higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE

LOGIC SYSTEMS N6

(8080306)

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09:00–12:00

Calculators may be used.

This question paper consists of 6 pages.

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DEPARTMENT OF HIGHER EDUCATION AND TRAINING
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE
LOGIC SYSTEMS N6
TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Write neatly and legibly.

NOTICE TO CANDIDATES

The lives and safety of people depend on logic systems. Logic systems form an integral part of intrinsic safety in equipment. Logic signalling systems prevent trains from colliding. Logic systems keep people safe in aircraft and in hospital intensive-care units. Your work, programming and designs, must be properly planned and presented at this diploma-level examination. Standards must be maintained for the safety of the public and safe operation of machinery.

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QUESTION 1: OPERATIONAL AMPLIFIERS

1.1 Operational amplifiers are still used extensively in instrumentation systems.

You are required to identify a fault within a robotic control loop. Within the circuitry is an inverting operational amplifier which is used to provide unity gain. All external connections are solid and correctly connected. Power supply voltages are correct at +15 V and -15 V. Voltages measured over the operational amplifier are as follows:

Input: +6 volts
Output: +6 volts

Explain in detail the reason why the operational amplifier must be replaced. (5)

1.2 Draw a large, neat, fully labelled circuit diagram of an operational amplifier configured as an R/2R digital-to-analogue converter. (10)

1.3 Draw the circuit of a noninverting op-amp and prove, by deriving the output equation, that the gain of the circuit will always be greater than one. (5)
[20]

QUESTION 2: INSTRUMENTATION

2.1 Draw a fully labelled block diagram of a staircase ramp analogue-to-digital converter. (10)

2.2 Draw a fully labelled block diagram of a typical character generator which will convert an ASCII character (for example 1000001) into a video display (the letter A) on a CRT (cathode-ray tube) screen. (10)
[20]

QUESTION 3: TRUE OR FALSE

Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'True' or 'False' next to the question number (3.1–3.10) in the ANSWER BOOK. Substantiate the answers.

3.1 Cellphone-signal multiplexing and switching is accomplished by logic systems.

3.2 The bus is the interface between the computer and an output peripheral like an LCD screen.

3.3 An input/output interfacing always incorporates an A/D or D/A conversion.

3.4 During the READ cycle, data is read from the accumulator.

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3.5 The instruction register holds only the operand address of the current instruction.

3.6 The program counter stores a count of all the programs stored in RAM.

3.7 Computer operations are synchronised with the system-clock oscillator.

3.8 Both machine code and assembler are classified as low-level languages.

3.9 The accumulator stores ongoing results of computations.

3.10 When a JUMP instruction is carried out, the program counter is loaded with the address of the instruction which is being jumped to.

(10 × 2) **[20]**

QUESTION 4: CIRCUITRY AND INTERFACING

4.1 Large-scale integration (LSI) is used to construct digital multimeters.

Use each of the following headings to discuss the integrated circuit functions and state whether they are analogue or digital, as well as where IC op-amps are used:

4.1.1 Input circuits

4.1.2 Processing circuits

4.1.3 Output circuits

(3 × 2) (6)

4.2 Draw a complete, fully labelled block diagram of a parallel ADC (flash converter) and list the decoding table for this converter. (6)

[12]

QUESTION 5: PROGRAMMING INSTRUCTION PROCEDURE

The following instruction set for a microprocessor is available:

OP-CODE	DESCRIPTION
LOAD	Load accumulator
ADDA	Add to accumulator
SUBT	Subtract from accumulator
MULT	Multiply
BRAN	Branch if the accumulator is negative
BRAP	Branch if the accumulator is positive
STOR	Store
OUTP	Output the contents of the accumulator
HALT	Halt (nonmemory referenced instruction)

Note the following specification information when writing the program:

- Show ALL mathematical operations done before coding.
- Case sensitivity: Operands must be lower case.
- Case sensitivity: Addresses must be upper case.
- Do NOT enlarge the instruction set.
- The program must stop processing if after any mathematical operation the contents of the accumulator are positive.
- Use the correct delimiter after delivering the output so that memory use is limited to program lines.
- Start the program at instruction location address AAAA.

The operands r, s, t, u v, w, x, y and z are stored in consecutive hexadecimal addresses:

Operand r is stored in address F1B1
Operand s is stored in address F1B2
Operand t is stored in address F1B3
Operand u is stored in address F1B4 and so on in sequence

The correct programming fields appear in the table below:

INSTRUCTION LOCATION	INSTRUCTION WORD		DESCRIPTION
	OP-CODE	ADDRESS	

Use the correct programming fields to write a program to calculate the value of x if:

x = tu + uu **[8]**

QUESTION 6: SIGNAL PROCESSING

6.1	6.1.1	Sketch a series D/A converter.	(5)
	6.1.2	Draw the input and output waveforms for the series D/A converter.	(2)
	6.1.3	Explain the operation of a series D/A converter.	(3)
6.2	6.2.1	Draw an operational-amplifier circuit which will function as a Schmitt trigger.	(3)
	6.2.2	Draw as input to the circuit in QUESTION 6.2.1 a 1 kHz sine wave synchronised on a timing diagram with the resultant output. Clearly show any upper and lower trip levels on the sine wave.	(4)
	6.2.3	Briefly explain the operation of the circuit in QUESTION 6.2.2.	(3)
			[20]

TOTAL: 100

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